



Graphing Rational Functions

Any function that can be written as the quotient of two polynomials can be called a **rational function**.

Finding the Domain of a Rational Function

Many rational functions have restricted domains because an expression is undefined if its denominator equals 0.

EXAMPLE 1

Determine the domain of the rational function.

A $f(x) = \frac{x + 5}{x^2 - 3x - 28}$

SOLUTION

Find domain restrictions by setting the denominator equal to 0.

$$\begin{aligned}x^2 - 3x - 28 &= 0 \\(x - 7)(x + 4) &= 0 \\x - 7 = 0 \quad \text{or} \quad x + 4 &= 0\end{aligned}$$

If $x = 7$ or $x = -4$, then the denominator equals 0. The domain is the set of all real numbers except 7 and -4 . You can state the domain in set notation.

$$\{x \in \mathbb{R}, x \neq 7 \text{ and } x \neq -4\}$$

B $g(x) = \frac{2}{x^2 + 1}$

SOLUTION

Find domain restrictions by setting the denominator equal to 0.

$$\begin{aligned}x^2 + 1 &= 0 \\x^2 &= -1\end{aligned}$$

The equation $x^2 = -1$ has no real solution, so there are no restrictions on the domain. The domain is the set of all real numbers. In set notation, the domain is $\{x \mid x \in \mathbb{R}\}$. ■

Recognizing the Rational Function Family of Graphs

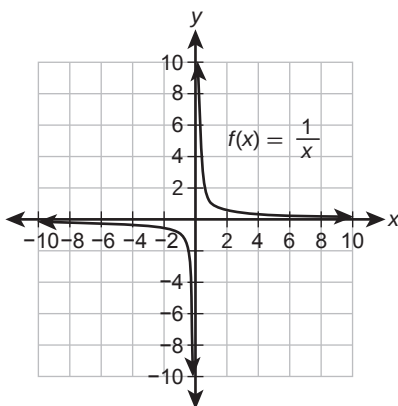
Equations of the form $f(x) = \frac{a}{x - h} + k$ are rational functions with one horizontal asymptote at $y = k$ and one vertical asymptote at $x = h$.

Rational Functions

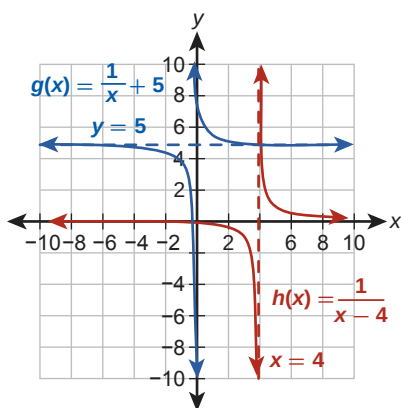
Rational Function Graph Family: $f(x) = \frac{a}{x-h} + k$

Each change in the value of a or b results in a different transformation of the parent graph,

$$f(x) = \frac{1}{x^n}.$$

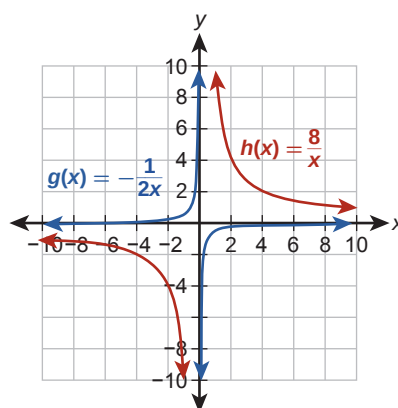


This graph is the graph of the parent function.



As h changes, the vertical asymptote shifts h units to the left or right.

As k changes, the horizontal asymptote shifts k units up or down.



If you change only the parameter a , you get curves that vary in their distance from the origin.

A negative a -value will reflect the graph across the x -axis.

Graphing Rational Functions

EXAMPLE 2

Graph the function.

A $y = \frac{1}{x+6}$

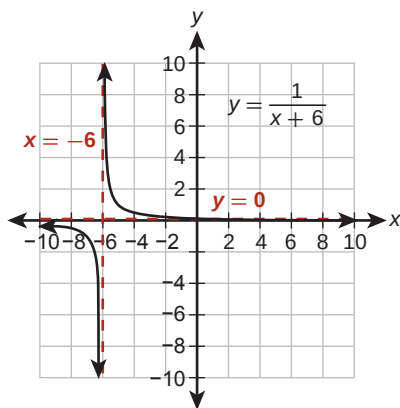
SOLUTION

► **Think About It** $\frac{1}{\text{little}} = \text{big}$
 $\frac{1}{\text{big}} = \text{little}$

If x is close to -6 , then $\frac{1}{x+6}$ is far from 0. If x is very big or very small, then

$\frac{1}{x+6}$ is close to 0. The domain of $y = \frac{1}{x+6}$ is $\{x \mid x \in \mathbb{R}, x \neq -6\}$.

The range of $y = \frac{1}{x+6}$ is $\{y \mid y \in \mathbb{R}, y \neq 0\}$. The line $x = -6$ is a vertical asymptote. The line $y = 0$ is a horizontal asymptote.

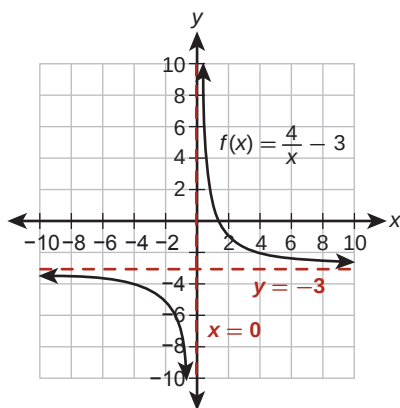


B $f(x) = \frac{4}{x} - 3$

SOLUTION

The domain of $f(x) = \frac{4}{x} - 3$ is $\{x \mid x \in \mathbb{R}, x \neq 0\}$ because the denominator cannot equal 0. The range of $f(x) = \frac{4}{x} - 3$ is $\{y \mid y \in \mathbb{R}, y \neq -3\}$ because $\frac{4}{x}$ can never be 0, so $\frac{4}{x} - 3$ can never be -3 .

► **Think About It** Any fraction $\frac{a}{b}$ equals zero if and only if a equals zero. So $\frac{4}{x}$ can never equal zero since 4 is not equal to zero.



Since 4 is greater than 1, the curve of the graph is farther away from the intersection of the asymptotes.

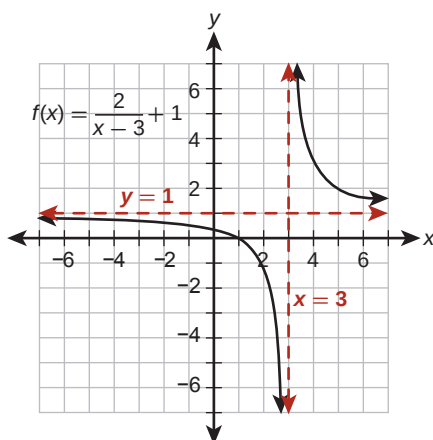
C $f(x) = \frac{2}{x-3} + 1$

SOLUTION

The domain is $\{x \mid x \in \mathbb{R}, x \neq 3\}$ because 3 is a zero of the denominator.

► **Think About It** A “zero of the denominator” is just a zero of the polynomial that is in the denominator.

The range of $f(x) = \frac{2}{x-3} + 1$ is $\{y \mid y \in \mathbb{R}, y \neq 1\}$ because $\frac{2}{x-3}$ can never be 0, so $\frac{2}{x-3} + 1$ can never be 1.



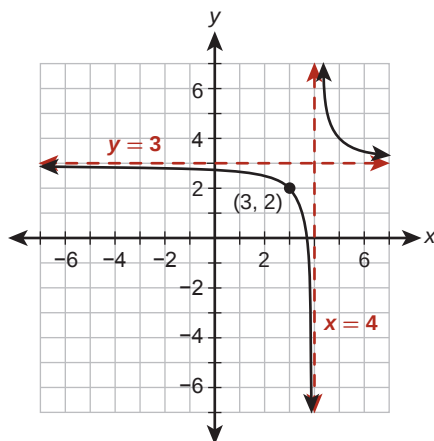
The lines $x = 3$ and $y = 1$ are asymptotes. ■

Finding the Equation When Given a Graph of a Rational Function in the Family $f(x) = \frac{a}{x - h} + k$

EXAMPLE 3

Find the equation in the family $f(x) = \frac{a}{x - h} + k$ for the graph of the function.

A



SOLUTION

Step 1 There is a horizontal asymptote at $y = 3$ and a vertical asymptote at $x = 4$. The equation of the graph so far is $f(x) = \frac{a}{x - 4} + 3$.

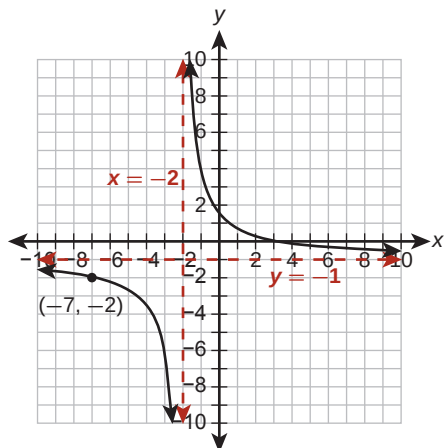
Step 2 Substitute the point $(3, 2)$ into the equation.

$$2 = \frac{a}{3 - 4} + 3 \quad \text{Substitute.}$$

$$2 = -a + 3 \quad \text{Simplify.}$$

$$a = 1 \quad \text{Solve.}$$

The equation of the graph is $f(x) = \frac{1}{x - 4} + 3$.

B**SOLUTION**

Step 1 There is a horizontal asymptote at $y = -1$ and a vertical asymptote at $x = -2$. The equation of the graph so far is $f(x) = \frac{a}{x+2} - 1$.

Step 2 Substitute the point $(-7, -2)$ into the equation.

$$-2 = \frac{a}{-7+2} - 1 \quad \text{Substitute.}$$

$$-1 = -\frac{a}{5} \quad \text{Simplify.}$$

$$5 = a \quad \text{Solve.}$$

The equation of the graph is $f(x) = \frac{5}{x+2} - 1$. ■

Graphing Rational Functions Not in the Family $f(x) = \frac{a}{x-h} + k$

EXAMPLE 4

Graph the function.

A $y = \frac{x^2 - 9}{x + 3}$

SOLUTION

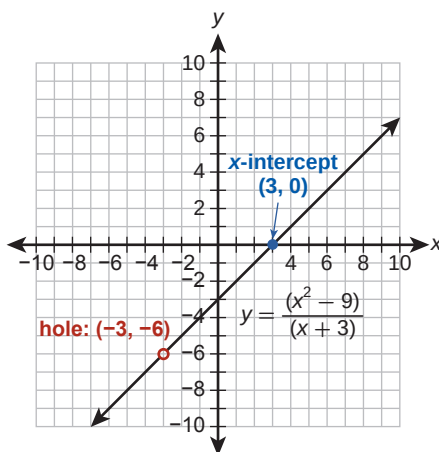
Factor and identify the zeros of the numerator and denominator.

$$y = \frac{x^2 - 9}{x + 3} = \frac{(x + 3)(x - 3)}{x + 3}$$

← The zeros are $x = -3$ and $x = 3$.
 ← The zero is $x = -3$.

For this function, -3 is a zero of both the numerator and the denominator, so there is a **hole at $x = -3$** . Meanwhile, 3 is a zero of only the numerator, so the **x -intercept is 3** , and $(3, 0)$ is on the graph. There are no zeros of only the denominator, so there is no vertical asymptote. The degree of the numerator is greater than the degree of the denominator ($2 > 1$), so there is no horizontal asymptote. If you divide out the common factor $x + 3$, you get the linear function $f(x) = x - 3$.

To graph $y = \frac{x^2 - 9}{x + 3}$, you can graph $f(x) = x - 3$, but you must put an open dot at $(-3, -6)$.



B $f(x) = \frac{1}{(x-3)^2}$

SOLUTION

If x is close to 3, then $\frac{1}{(x-3)^2}$ is far from 0. If x is far from 3,

then $\frac{1}{(x-3)^2}$ is close to 0. If $x = 3$, then the denominator is zero

and the function is undefined, so the domain of $f(x) = \frac{1}{(x-3)^2}$ is

$\{x \mid x \in \mathbb{R}, x \neq 3\}$. The numerator is positive and the denominator

is a squared value, so it has to be positive as well, so the range is

$\{y \mid y \in \mathbb{R}, y > 0\}$. Because $(x-3)$ is in the denominator, there is one

vertical asymptote at $x = 3$. The degree of the numerator is 0 and the degree

of the denominator is 2. Because $0 < 2$, the x -axis, $y = 0$, is the horizontal

asymptote.

